



PROCESSAMENTO DE BIG DATA

Presidenciais 2026

Grupo 09

DADOS ANALISADOS

Detected People

Emotion Prediction

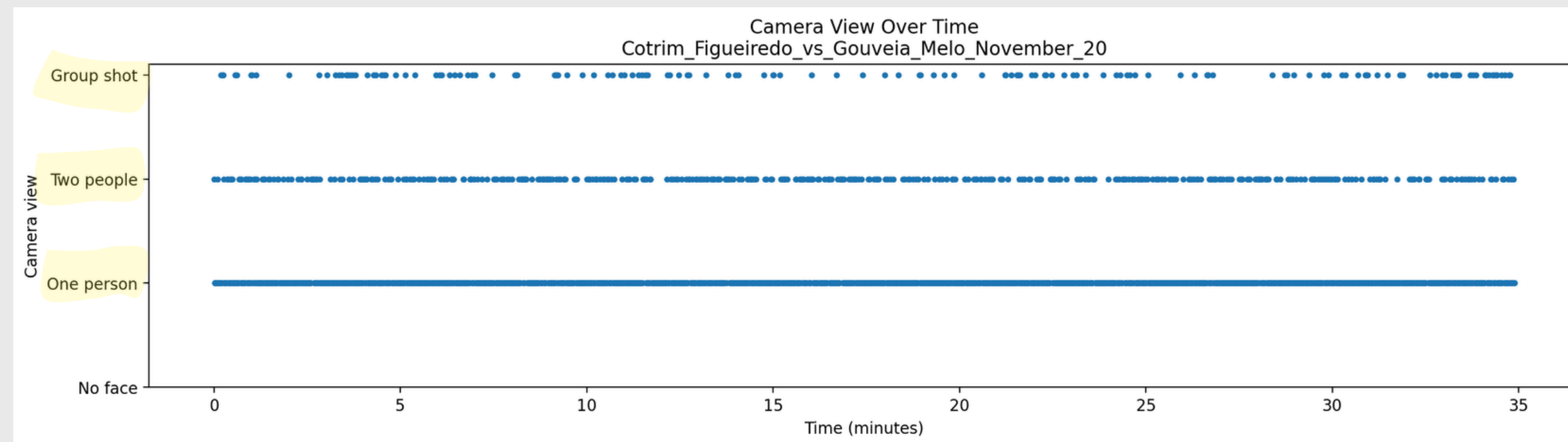
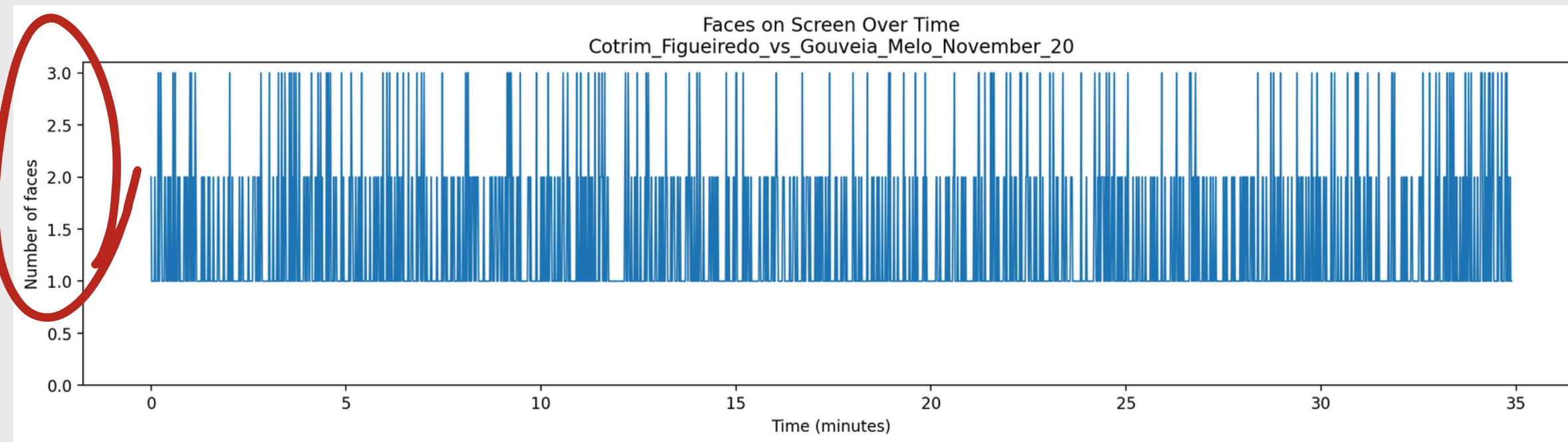
**Body Pose
Keypoints**

**Emotion Probability
Vector**

Detected Faces

Facial Landmarks

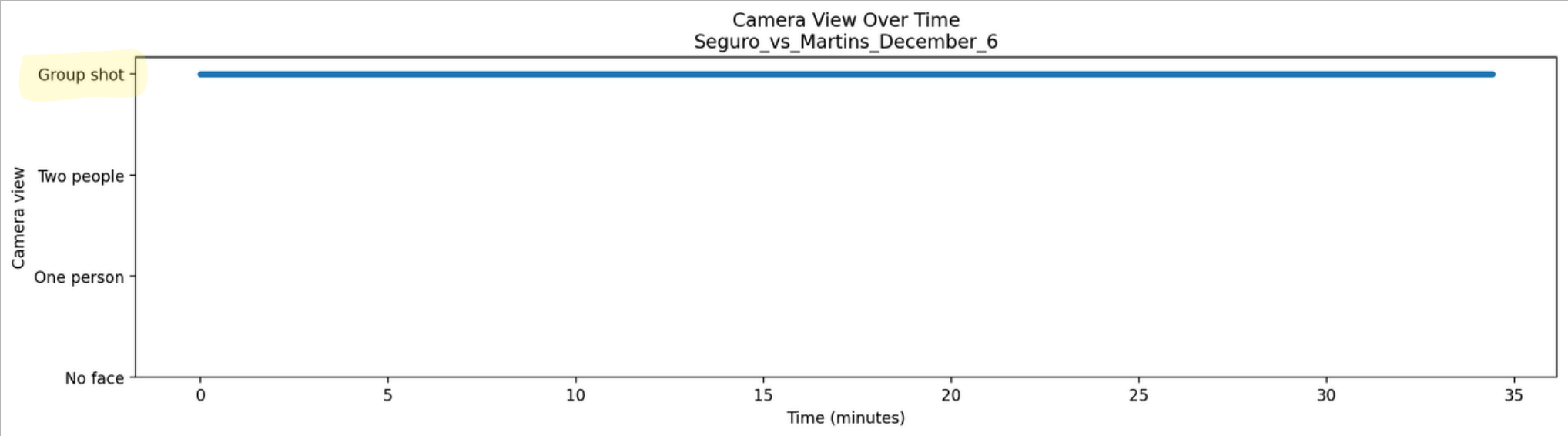
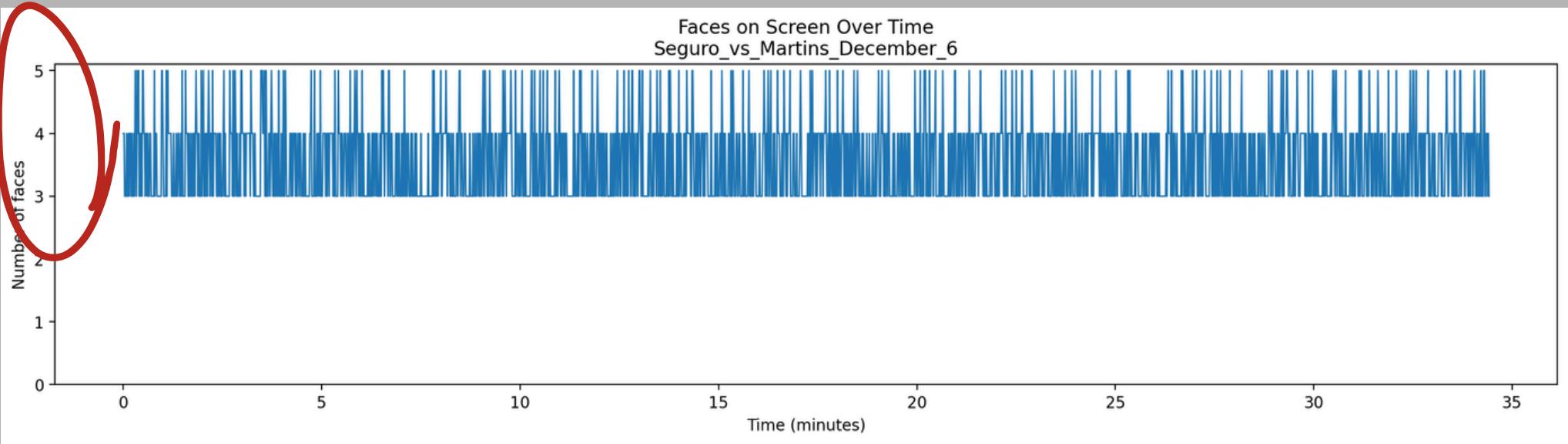
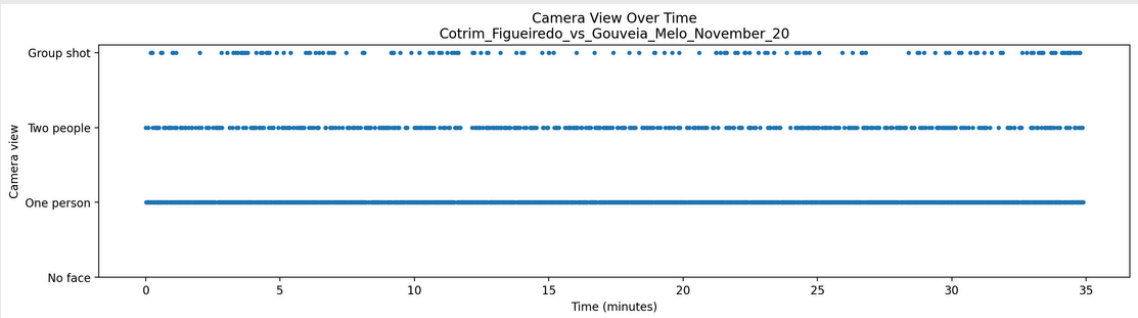
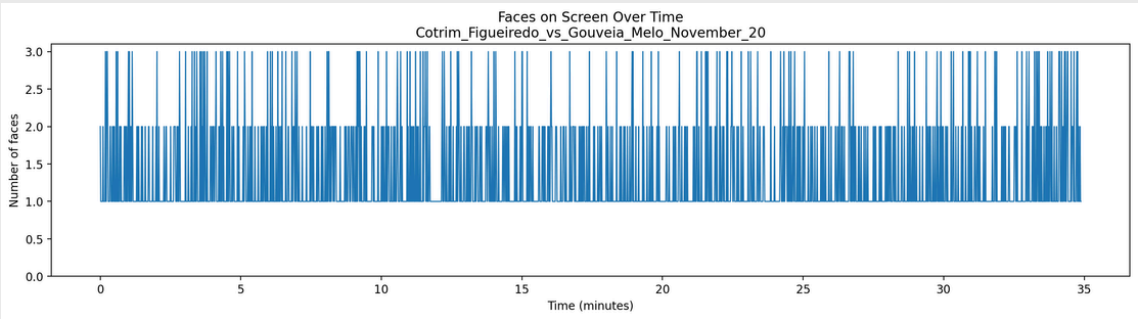
DEBATE NORMAL



RESULTADOS INICIAIS

ERROS ENCONTRADOS

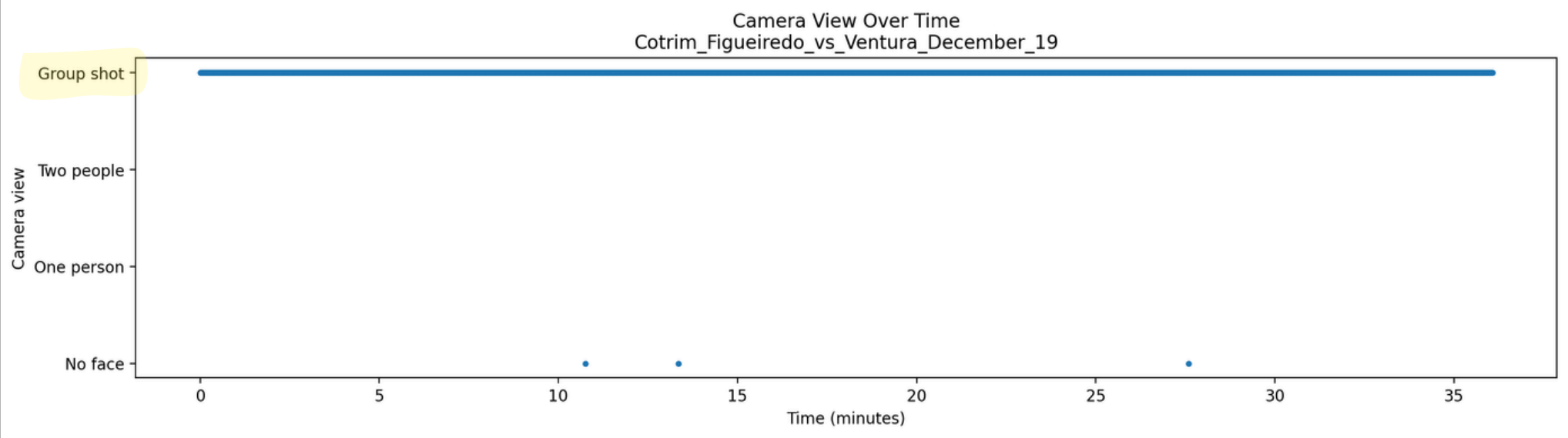
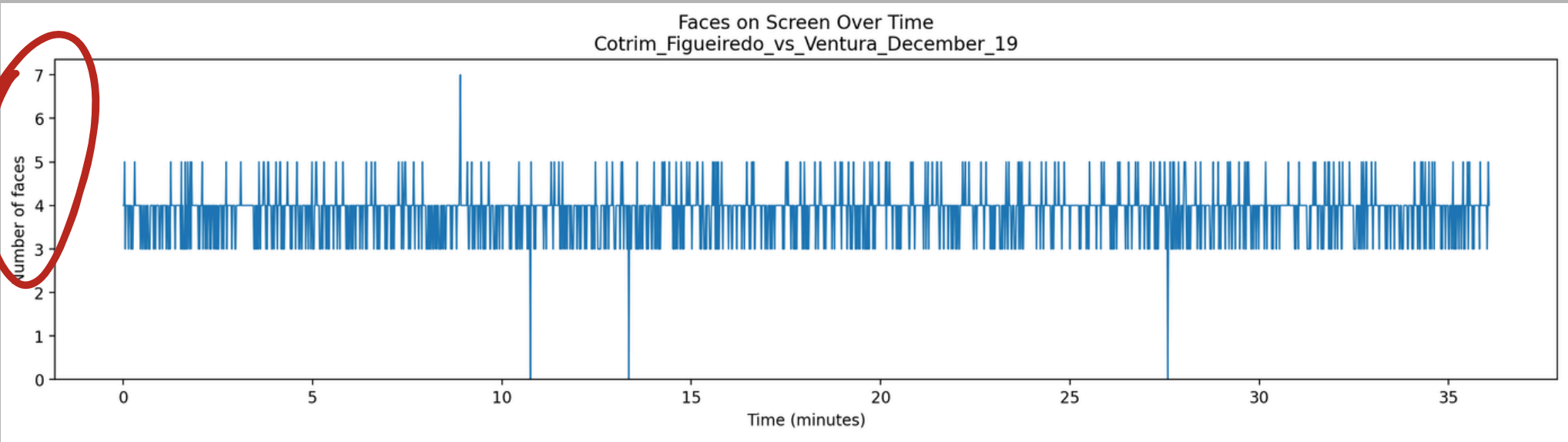
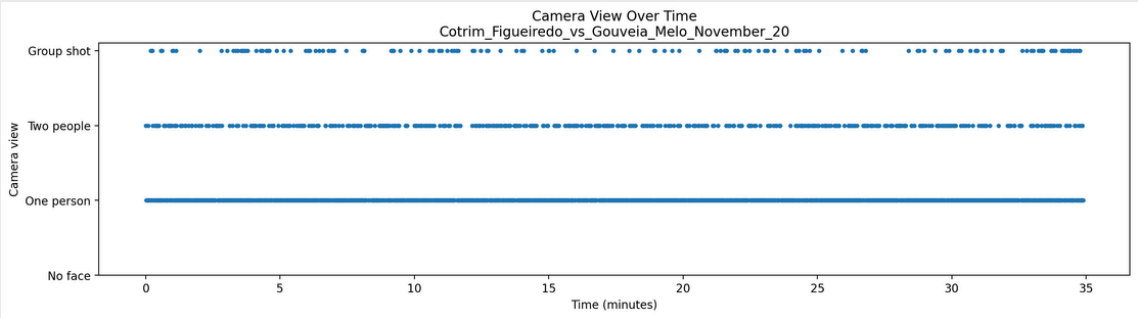
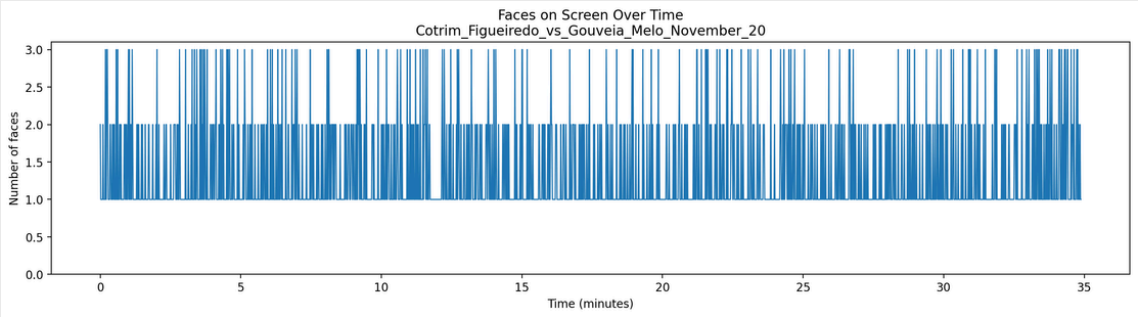
DEBATE NORMAL



RESULTADOS INICIAIS

ERROS ENCONTRADOS

DEBATE NORMAL



ERRO

- DUAS CAIXAS NA MESMA PESSOA
- DETEÇÃO DE PESSOA EXTRA

CORREÇÃO

Frame index 16 — Frames/Cotrim_Figueiredo_vs_Marques_Mendes_December_7/frame_1422.jpg

RAW pickle detections
people=3 | faces=2



CLEANED detections
people=1 | faces=1



ERRO

- DUAS CAIXAS NA MESMA PESSOA
- DETEÇÃO DE PESSOA EXTRA

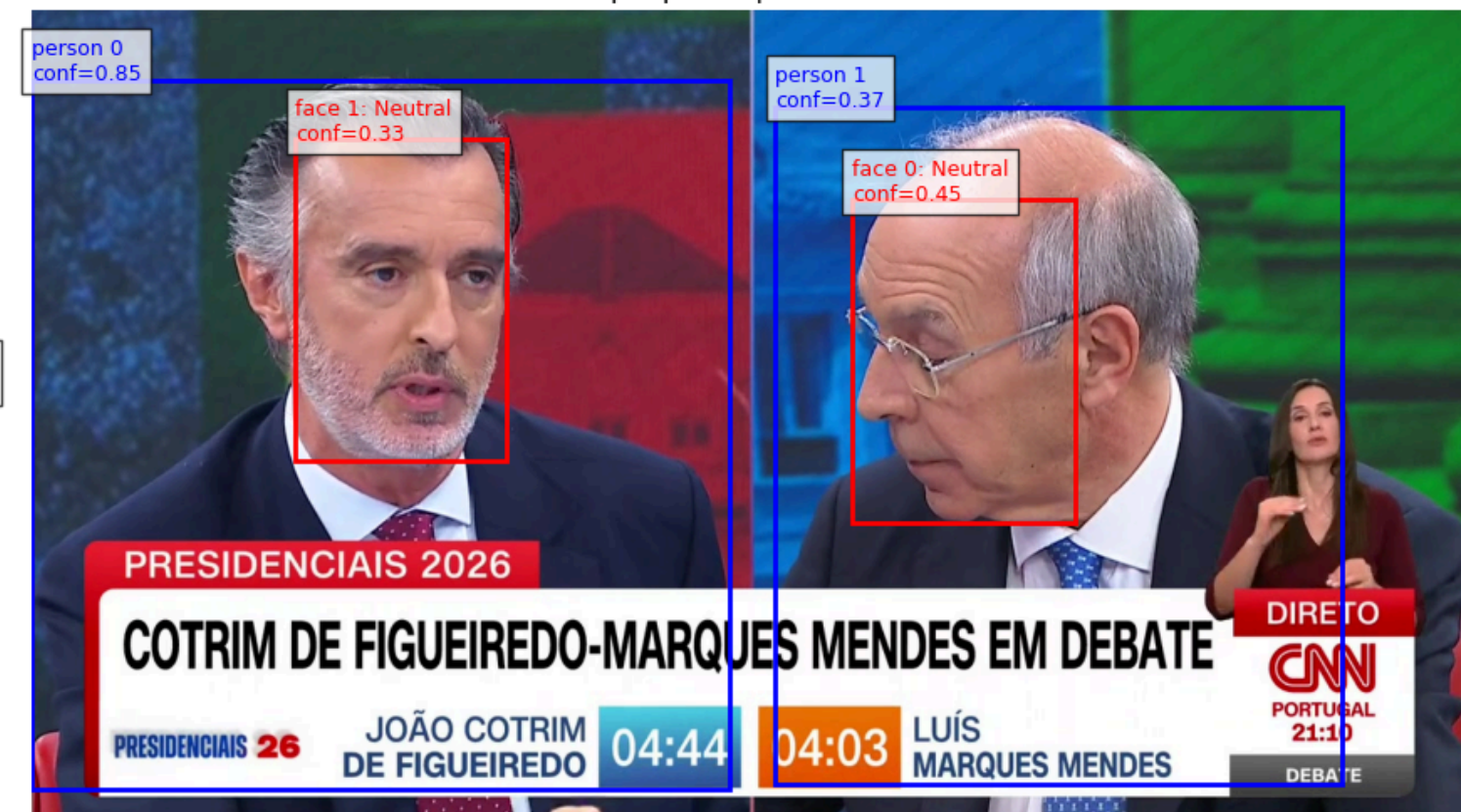
CORREÇÃO

Frame index 11 — Frames/Cotrim_Figueiredo_vs_Marques_Mendes_December_7/frame_650.jpg

RAW pickle detections
people=6 | faces=3



CLEANED detections
people=2 | faces=2



ERRO

- DETEÇÃO DE PESSOAS EXTRA

CORREÇÃO

Frame index 4 — Frames/Cotrim_Figueiredo_vs_Ventura_December_19/frame_1625.jpg

RAW pickle detections
people=4 | faces=4



CLEANED detections
people=2 | faces=2



ERRO

- DETEÇÃO DE PESSOAS EXTRA

CORREÇÃO

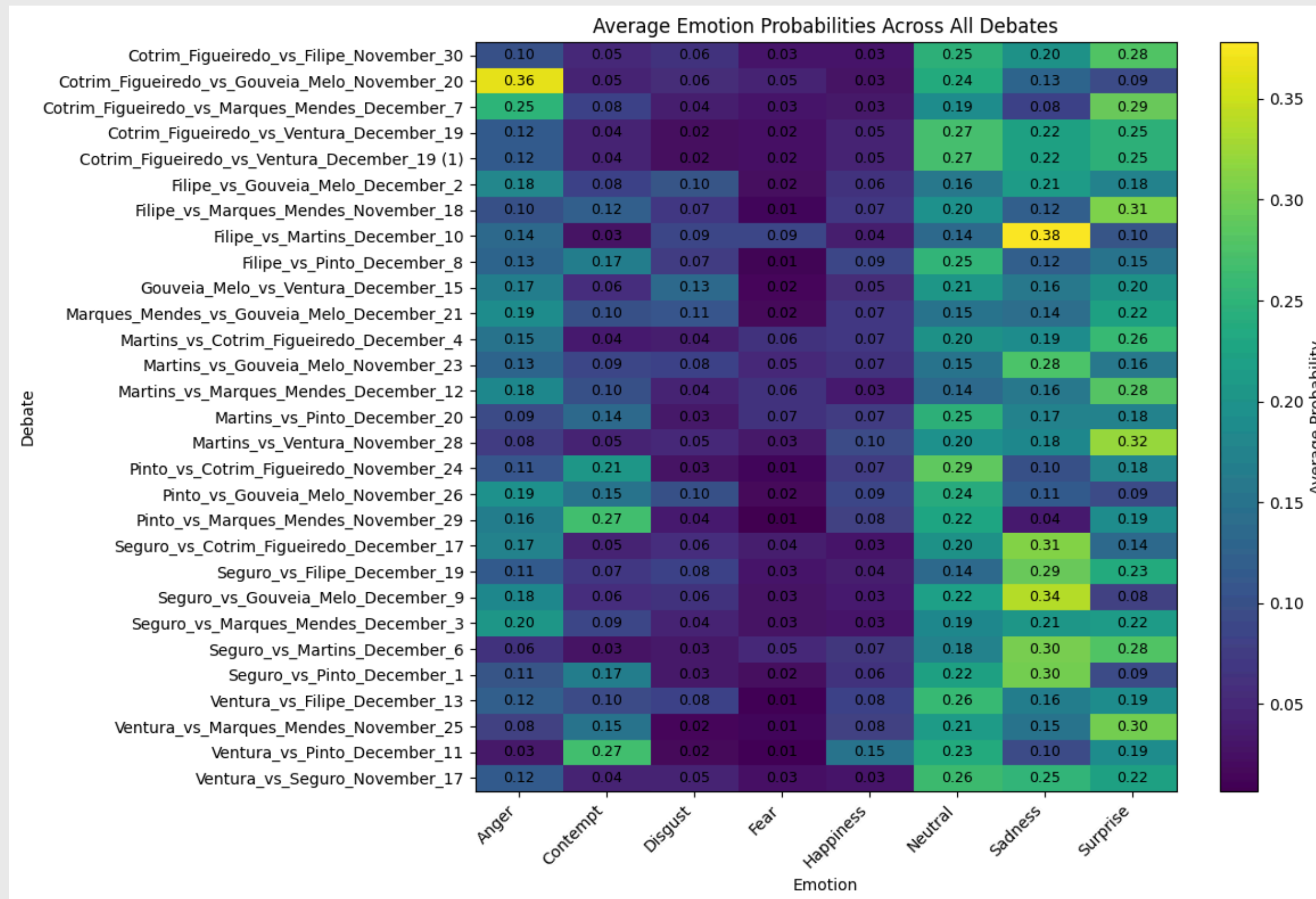
Frame index 0 — Frames/Seguro_vs_Martins_December_6/frame_1956.jpg

RAW pickle detections
people=4 | faces=4

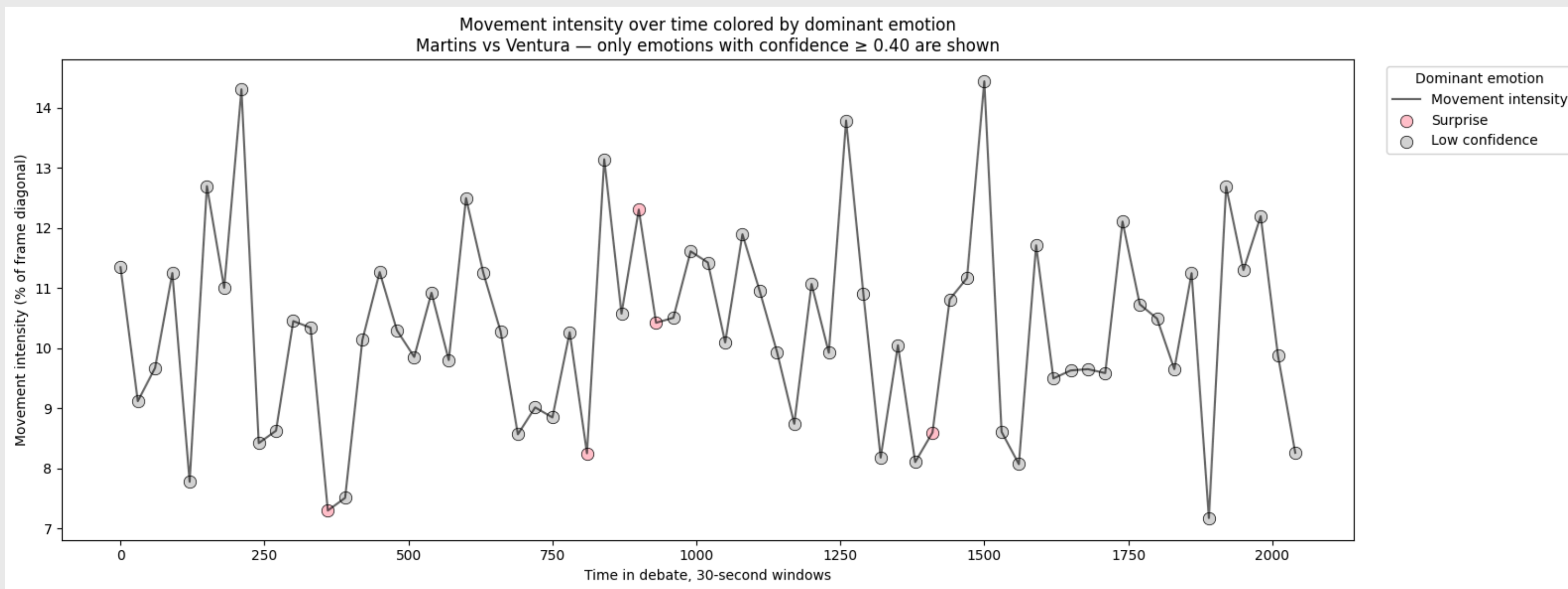


CLEANED detections
people=2 | faces=2

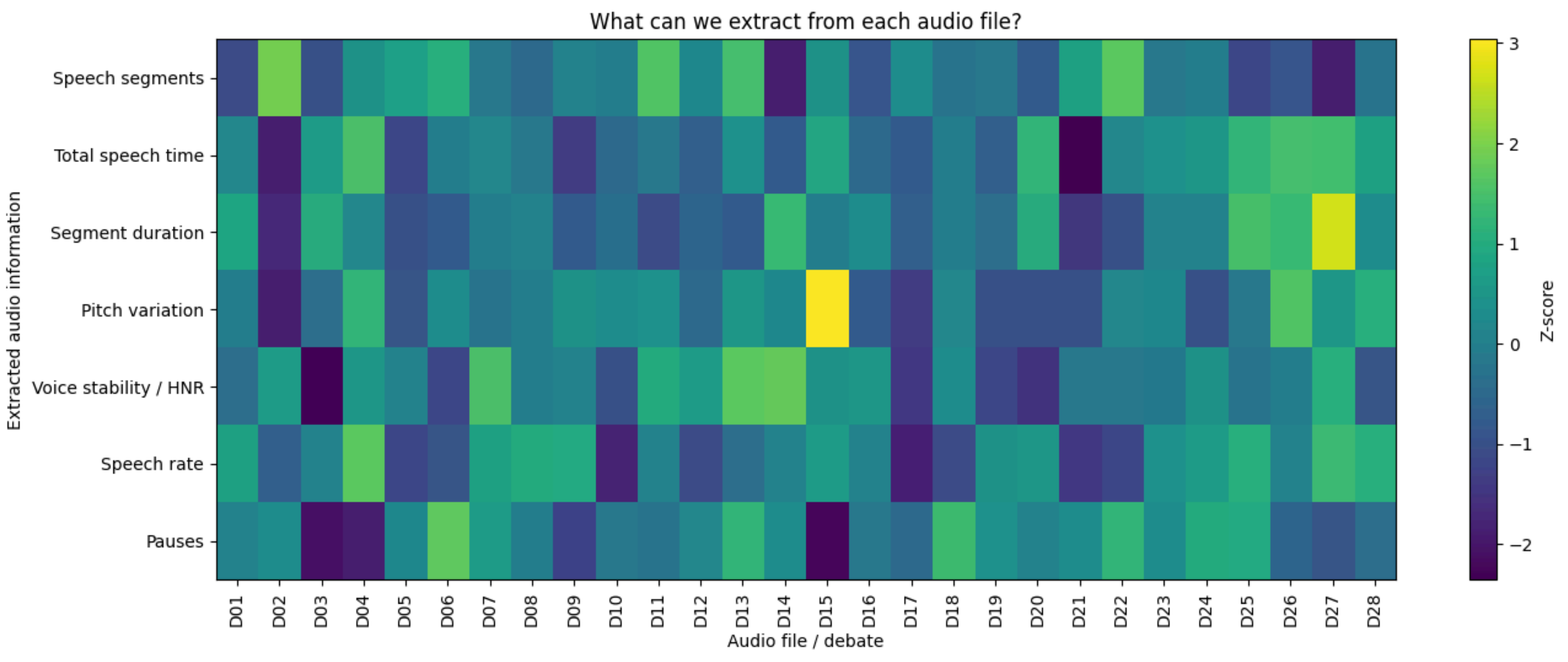




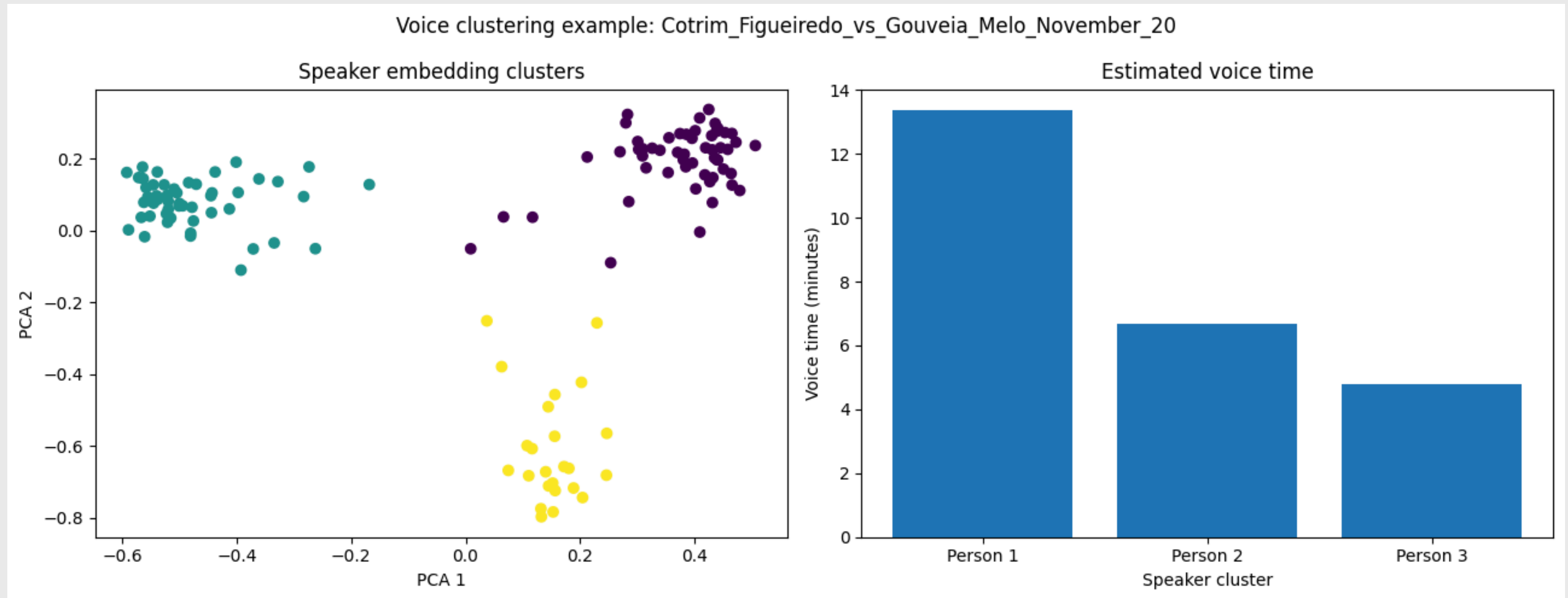
INTENSIDADE DE MOVIMENTO



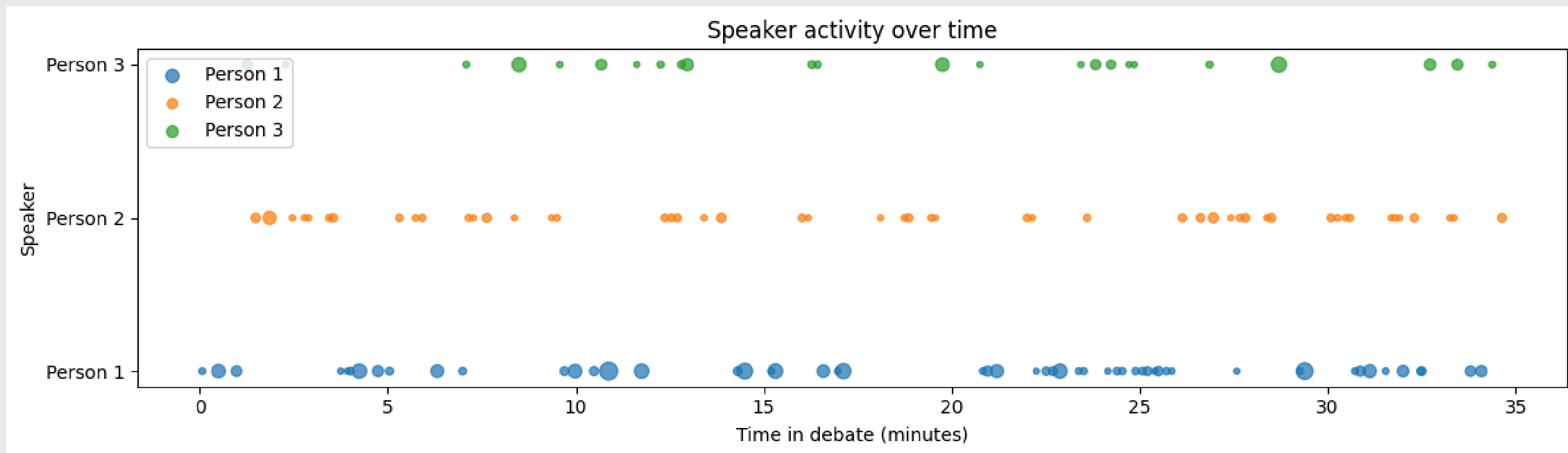
Audio files: what can we extract?



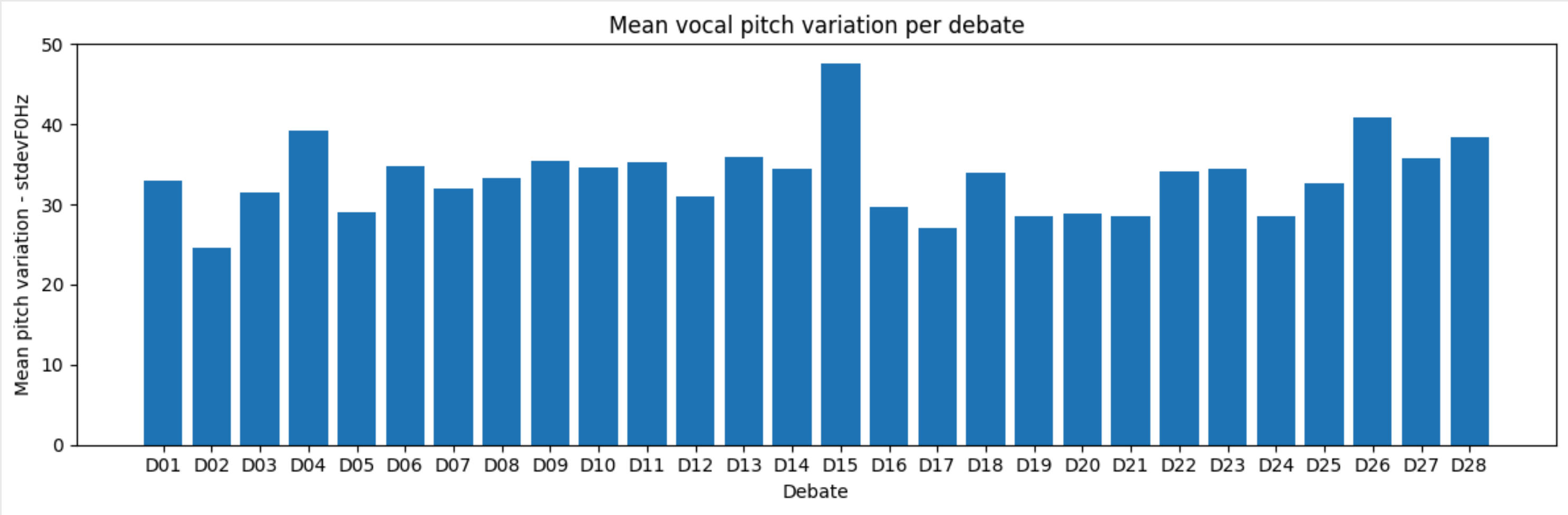
Speaker embeddings: identifying who is speaking



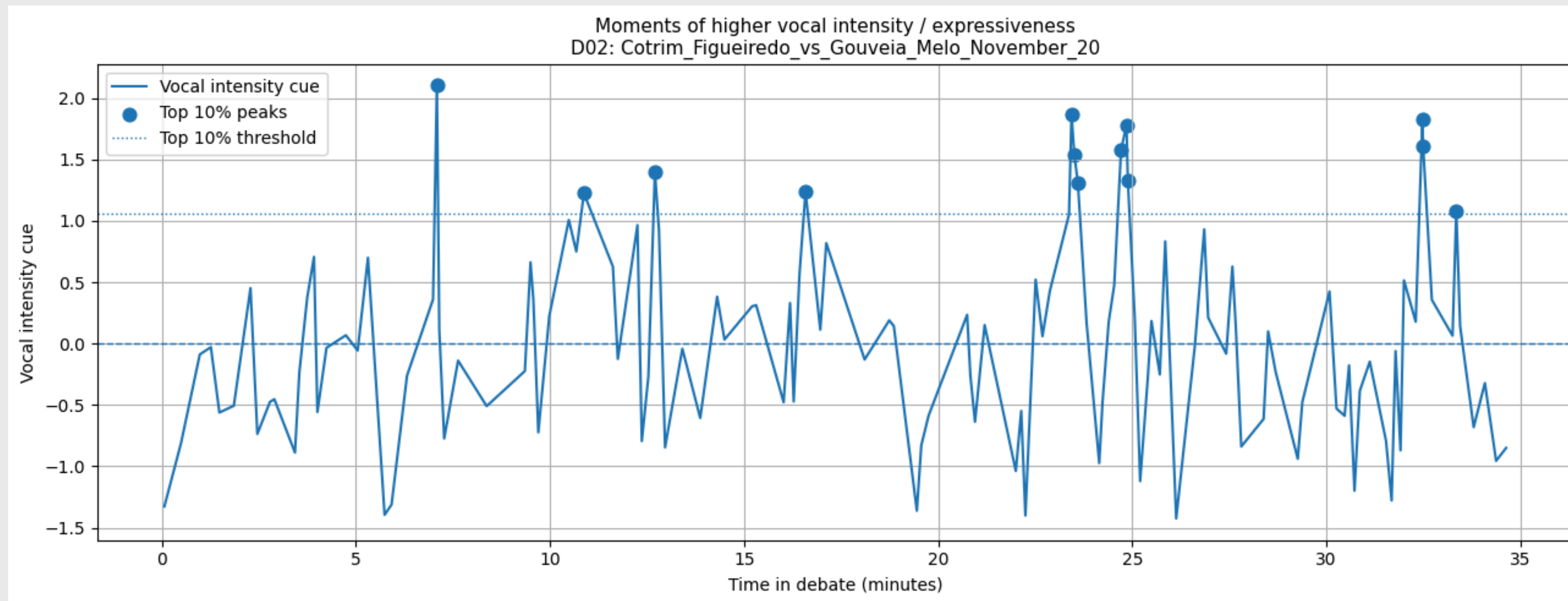
Speaker embeddings: identifying who is speaking and when



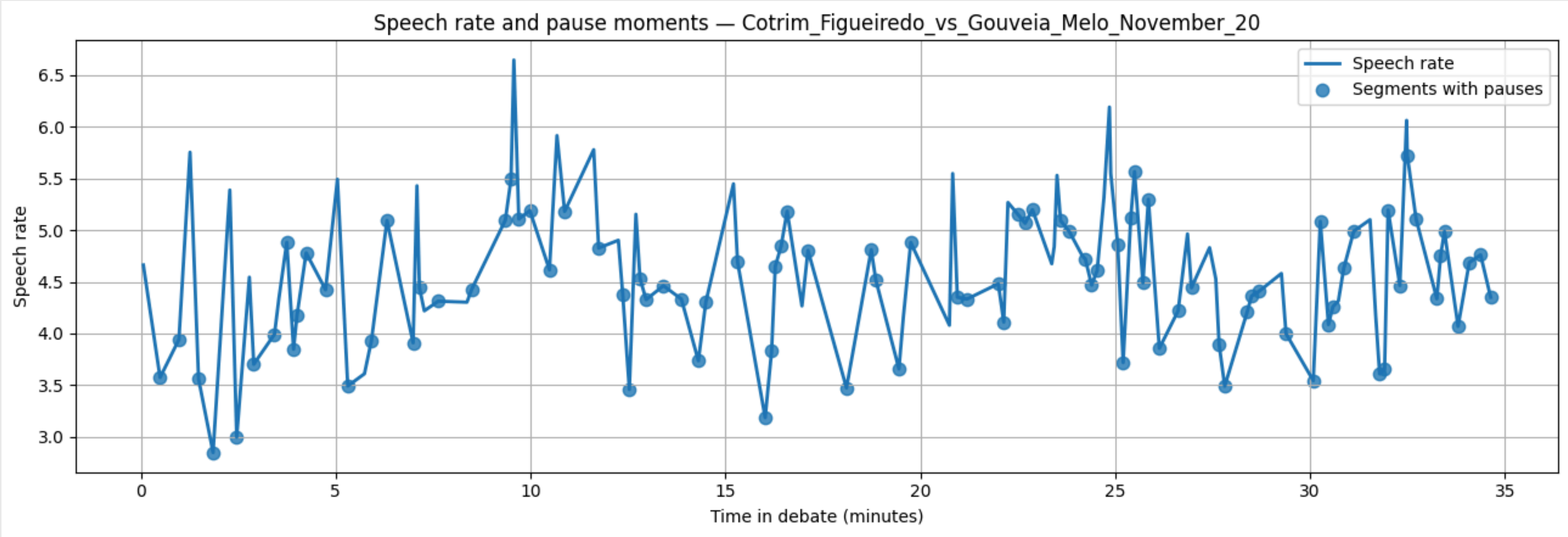
Variation in vocal intensity: a possible cue for emotional expression



Variation in vocal intensity: a possible cue for emotional expression



Speech rate and pause





Next step: linking audio with other files

[AUDIO.PKL]

speaker embeddings

speech rate + pauses

vocal intensity

pitch variation



[POSSIBLE EMOTIONAL CUES]

expressiveness

hesitation

intense response

possible tension



[VISUAL.PKL] + [SPEECH.PKL]

facial emotions transcript

screen presence spoken content

SPEECH (DEBATES E TELEJORNAIS)

28

Debates



843

Segmentos de texto

14

Telejornais



564

Segmentos de texto



Timestamp

Localização temporal
precisa do início de cada
intervenção no áudio.



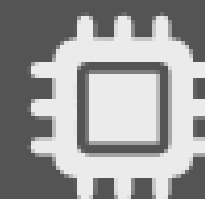
Duration

Tempo de exposição
contínua de cada
segmento

“

Transcript

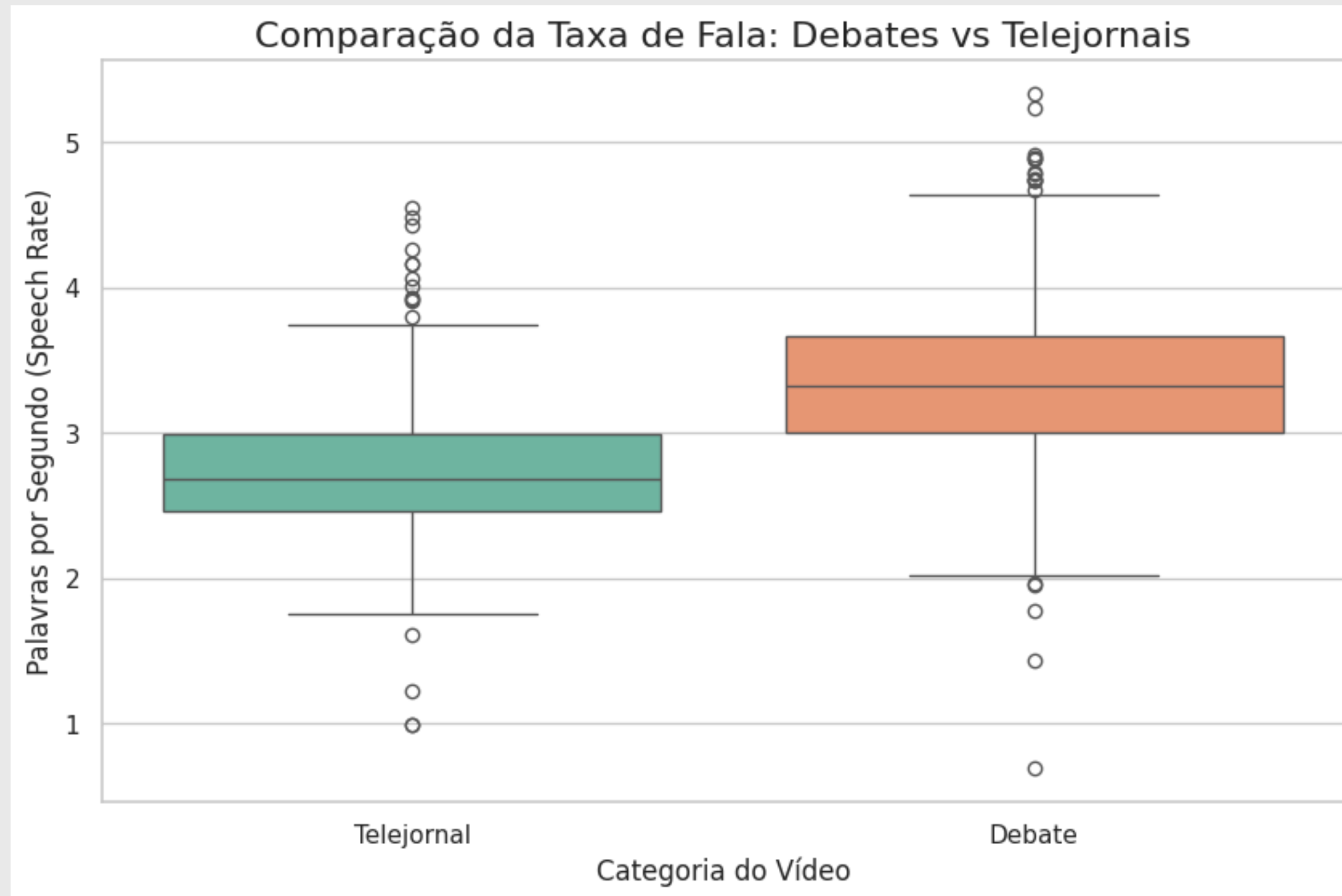
Conteúdo textual extraído
via OpenAI Whisper



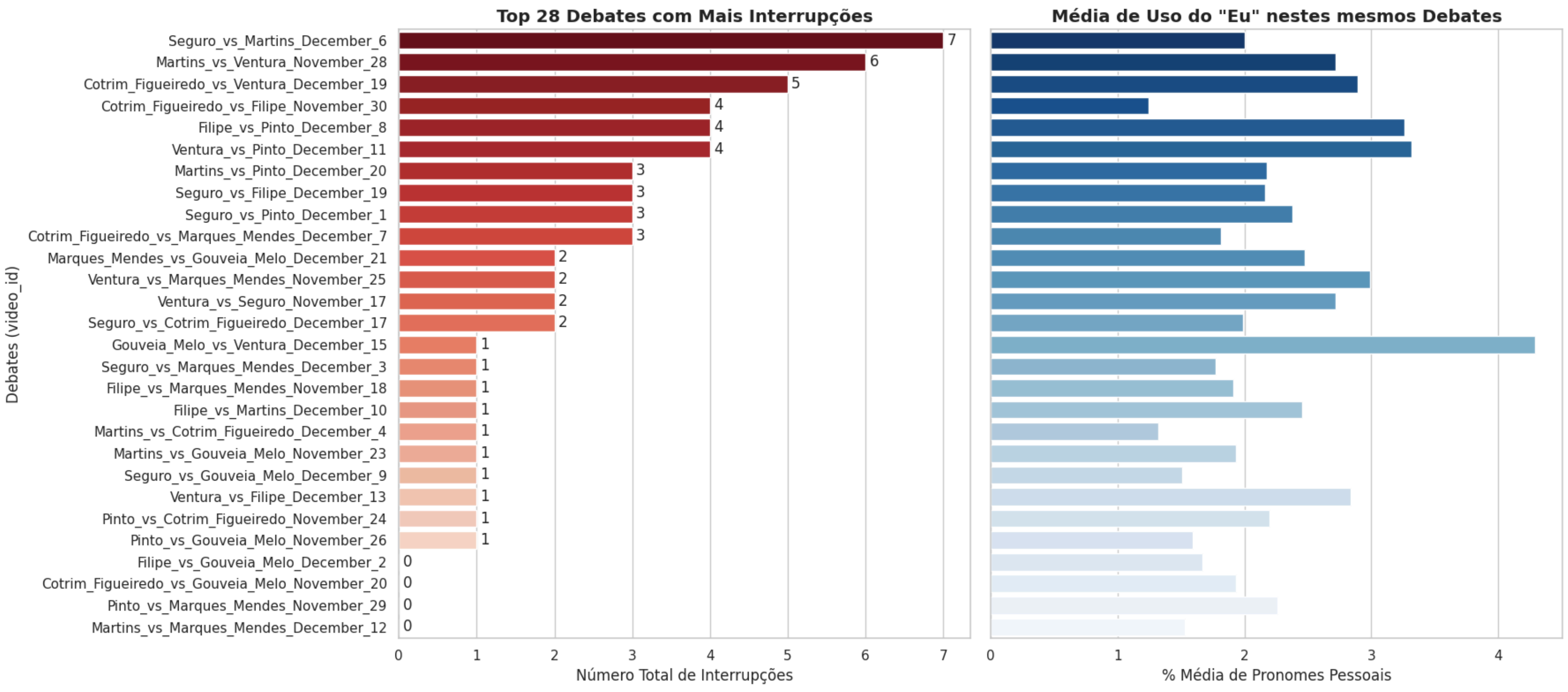
Embeddings

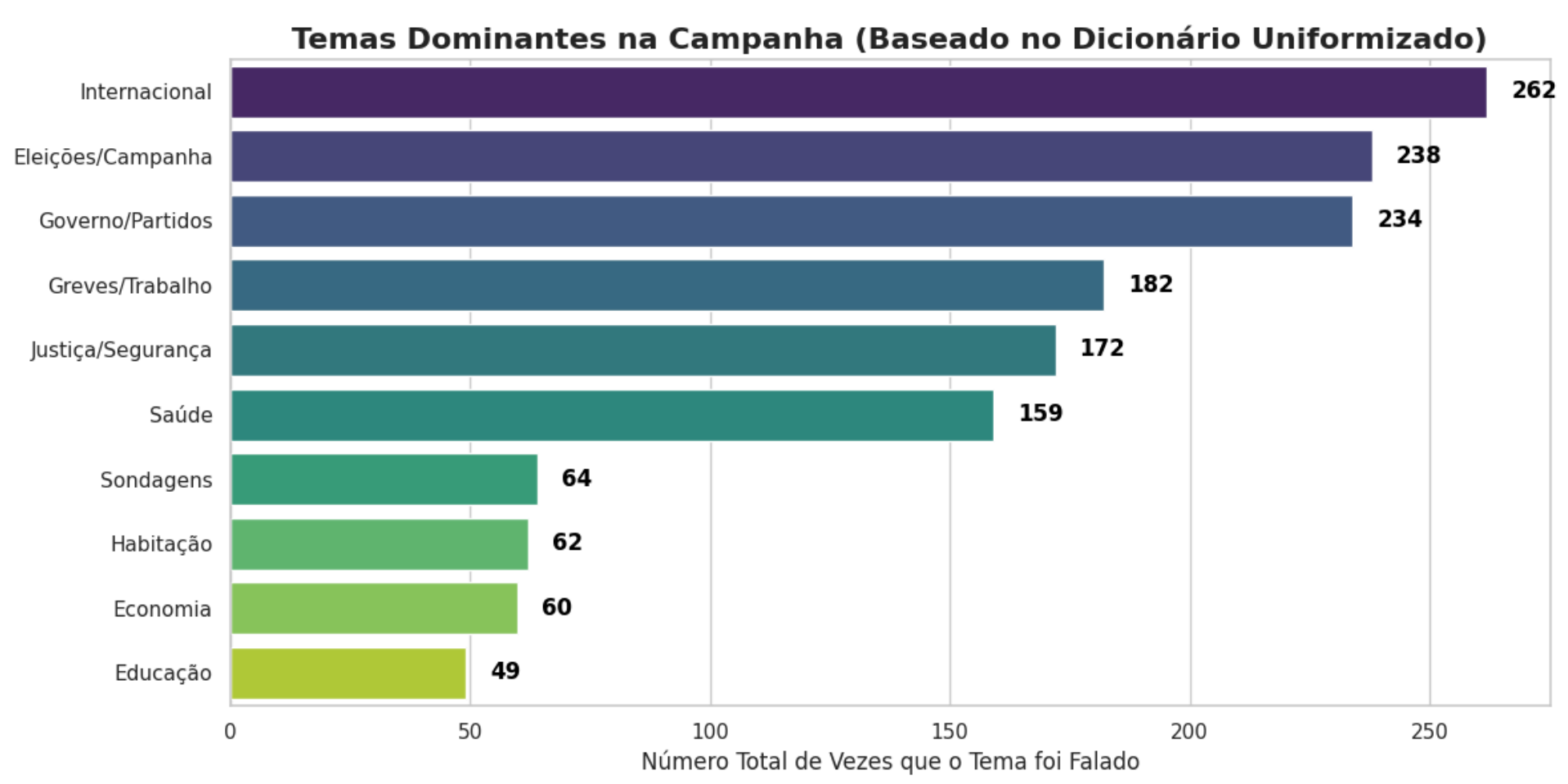
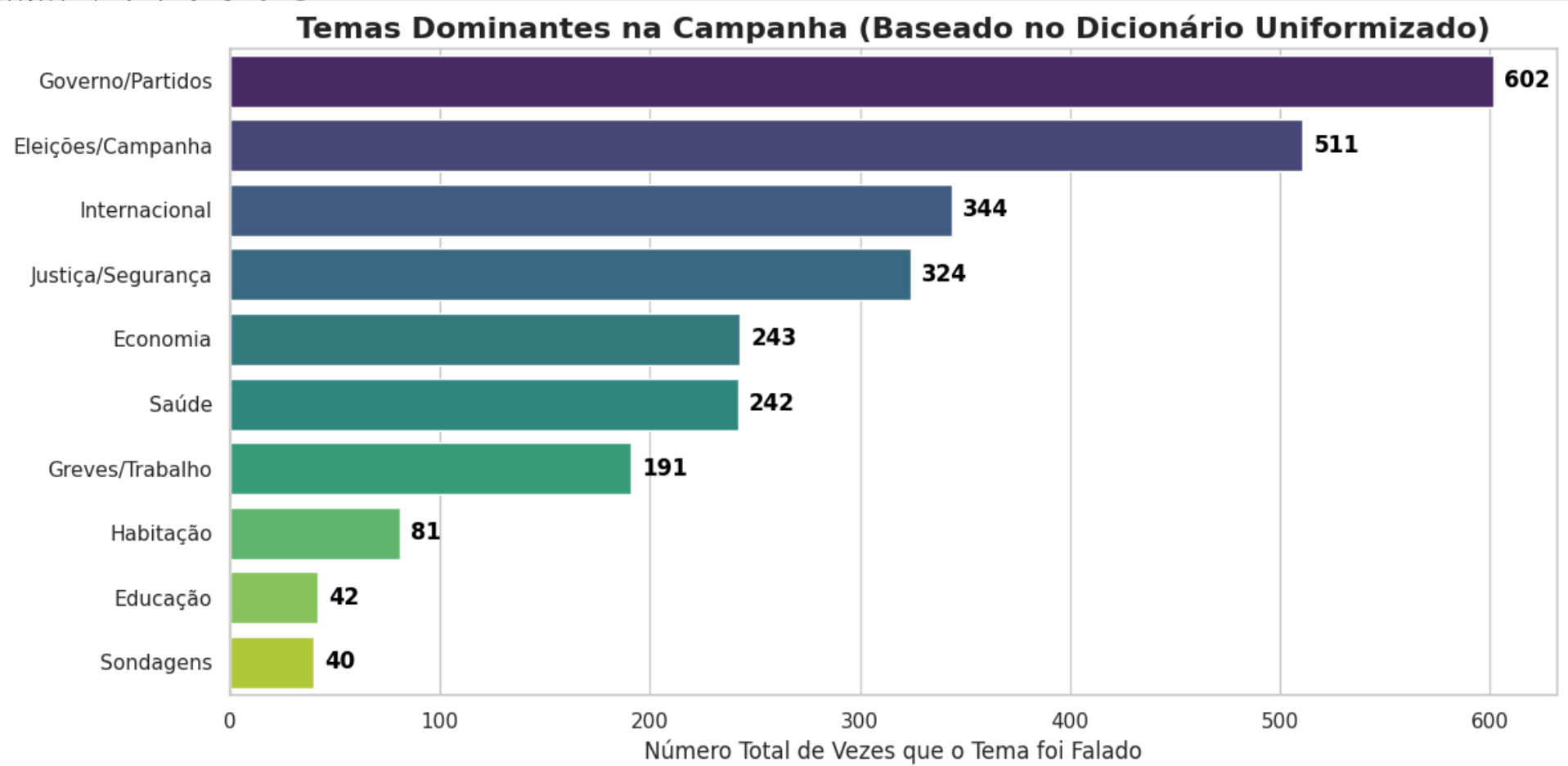
Representação vetorial do
texto para análise.

SPEECH RATE: DEBATES VS. NOTÍCIAS



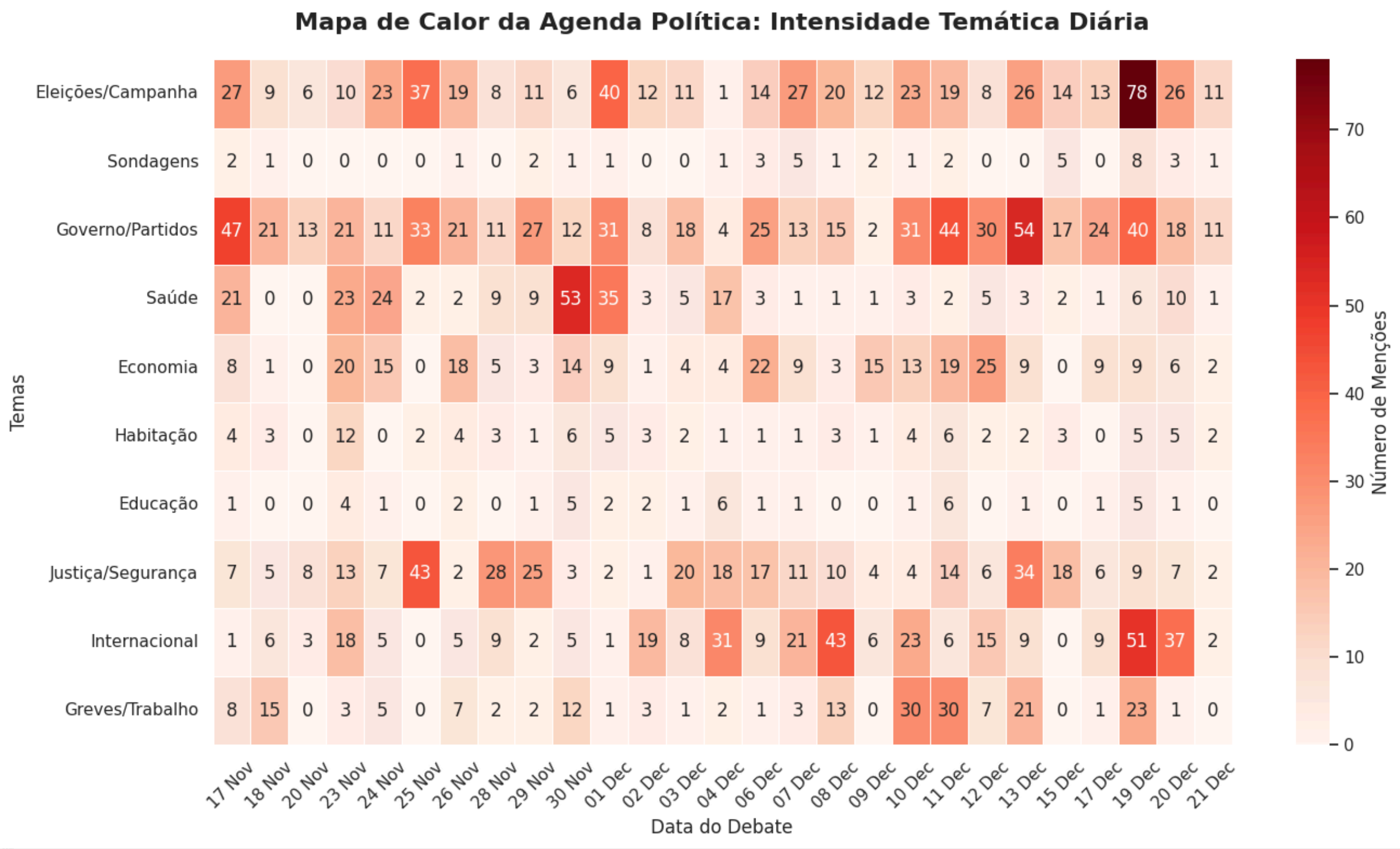
ANÁLISE DEBATES: PERSONALIZAÇÃO E INTERRUPÇÕES



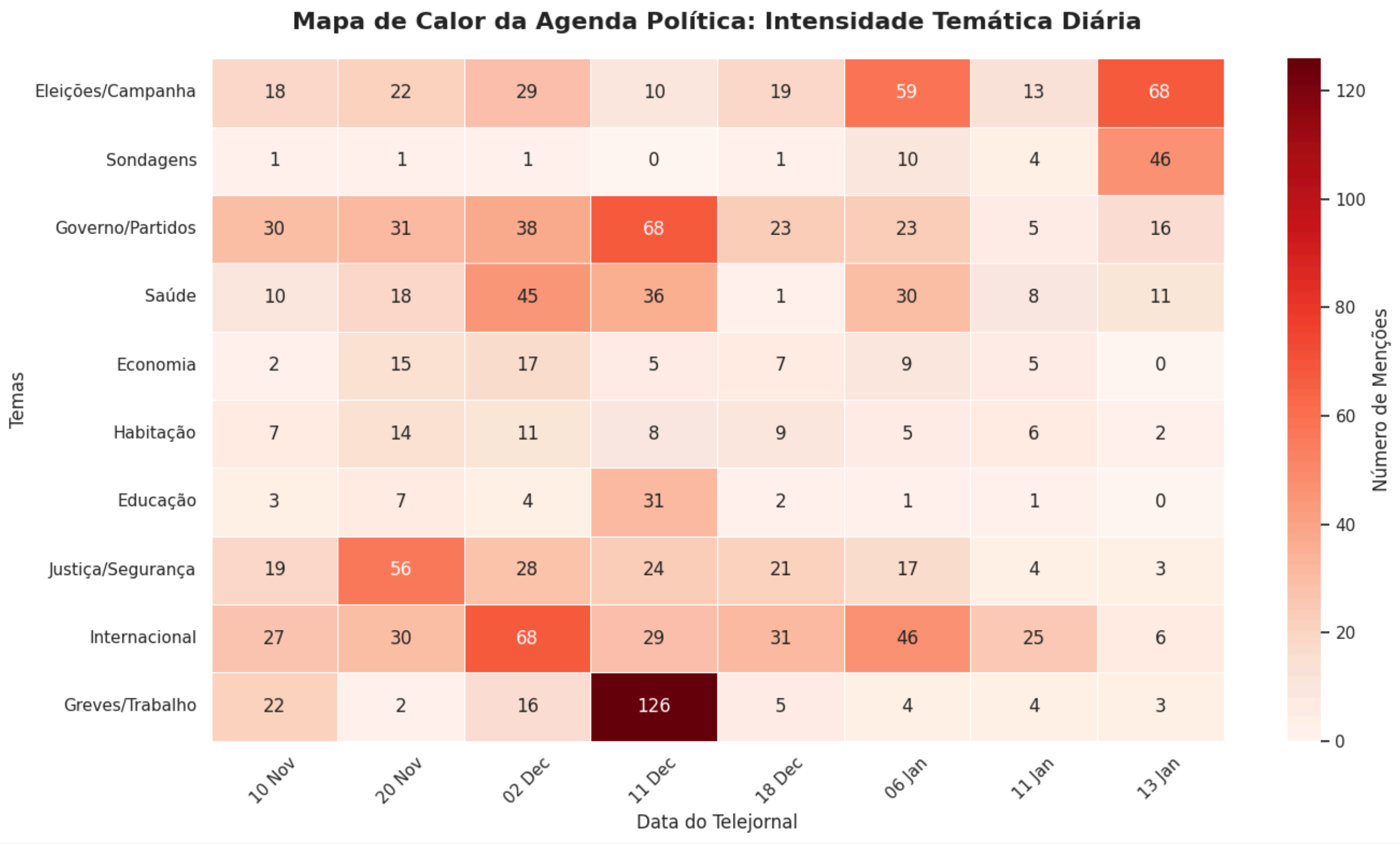


ANÁLISE TEMÁTICA DEBATES VS NOTÍCIAS

EVOLUÇÃO TEMPORAL DE TEMAS (DEBATES)

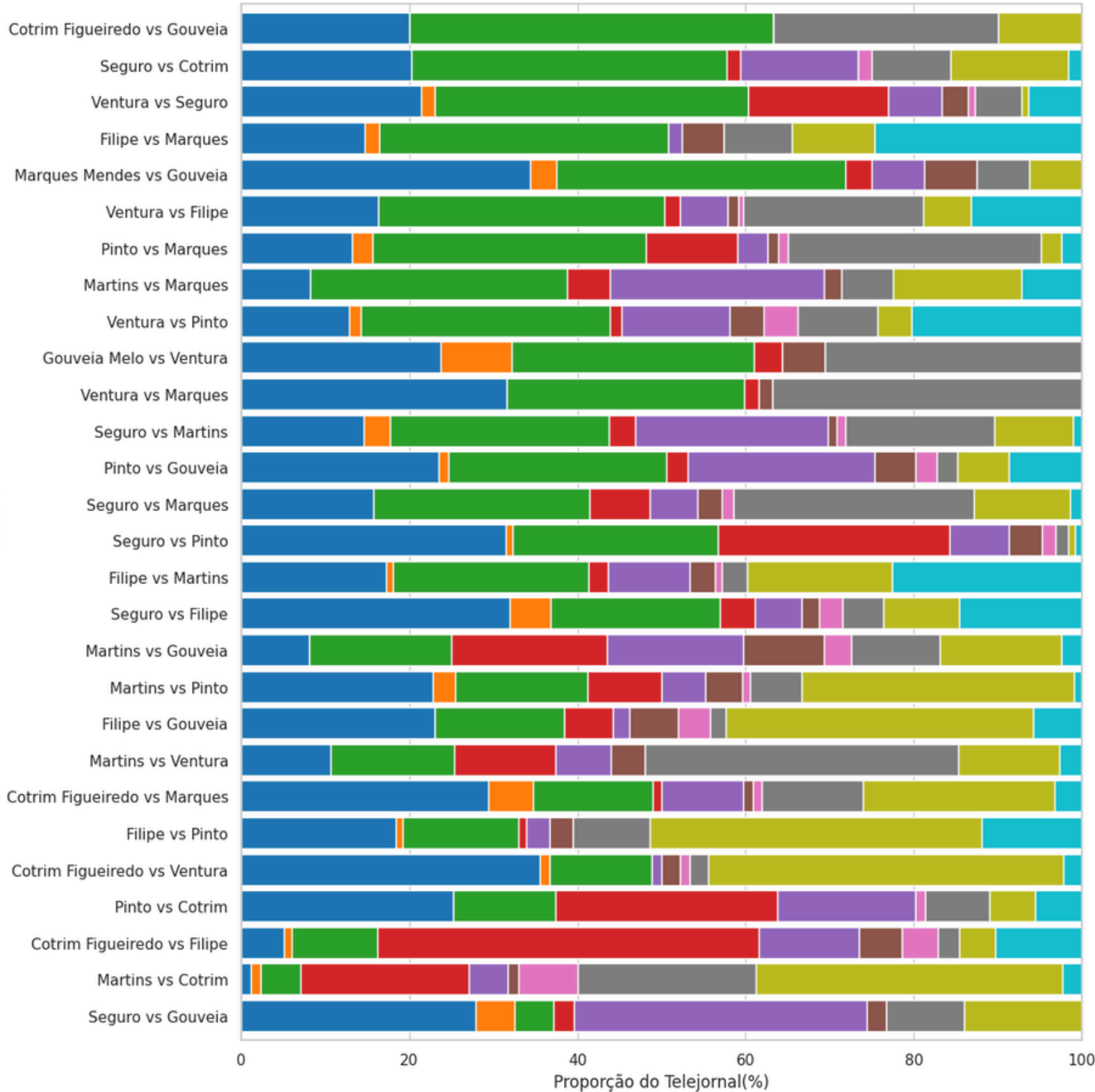


EVOLUÇÃO TEMPORAL DE TEMAS (NOTÍCIAS)

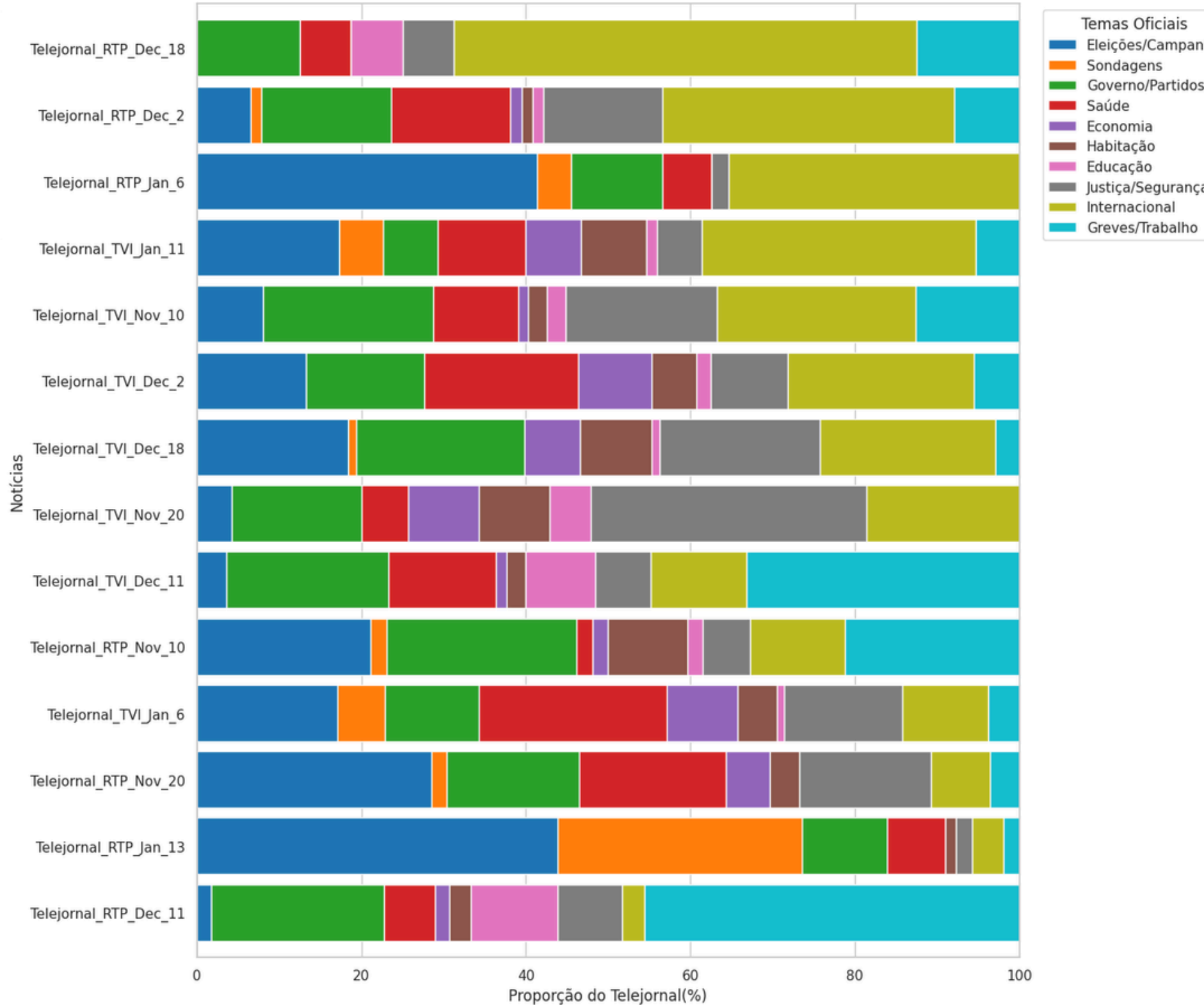


PROPORÇÃO TEMÁTICA

Temas nos Debates



Temas no telejornal



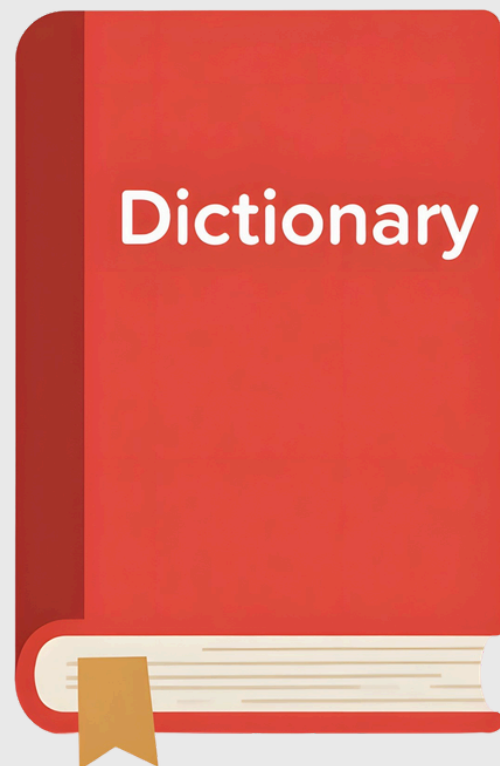
OCR: DICIONÁRIO DE REFERÊNCIAS

Para cada candidato, partido ou tema, criamos uma lista de formas possíveis como a referência pode aparecer no OCR.

Candidatos

Temas

Exemplos:



Exemplo: André Ventura

André Ventura
ventura
líder do Chega
presidente do Chega

Economia
Educação
Eleições/Campanha
Governo/Partidos
Greves/Trabalho
Saúde
Habitação
Internacional
Justiça/Segurança
Sondagens

“saúde”
“SNS”
“hospital”
“hospitais”
“médico”
“médicos”
“enfermeiro”
“urgência”
“ambulância”
“lista de espera”

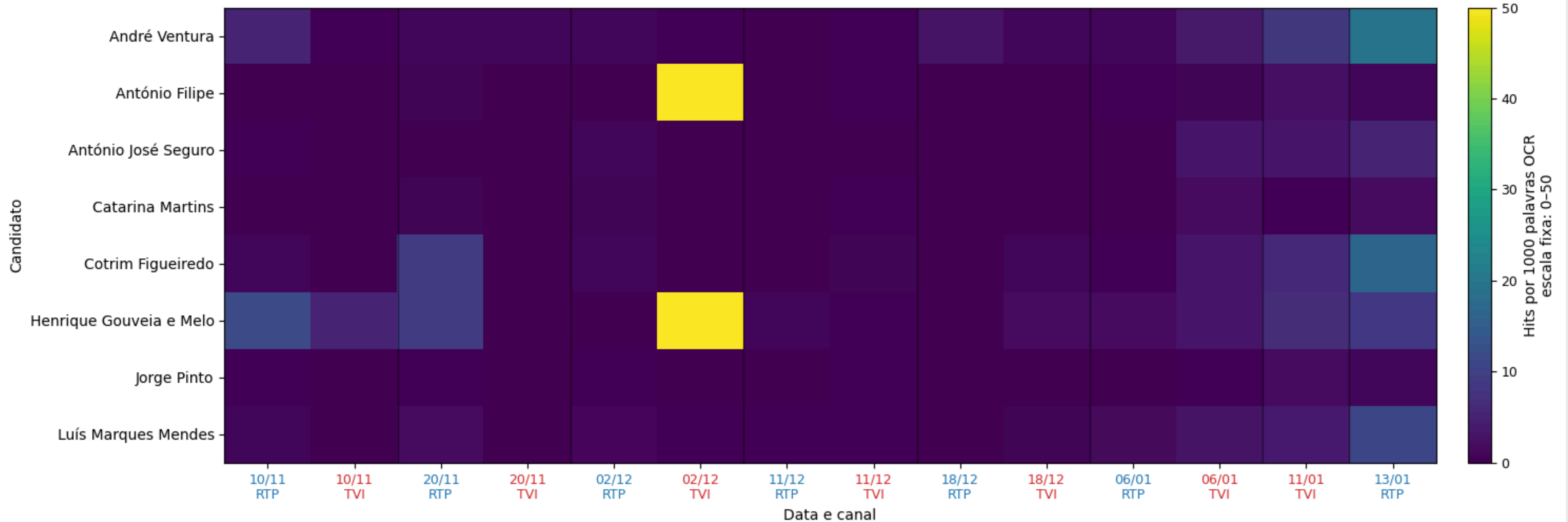


Objetivo: detetar menções mesmo quando o OCR não escreve sempre o nome da mesma forma.

ANÁLISE DOS TELEJORNAIS

OCR

Candidatos por telejornal — normalizado por 1000 palavras OCR





TELEJORNAL_TVI_DEC_2

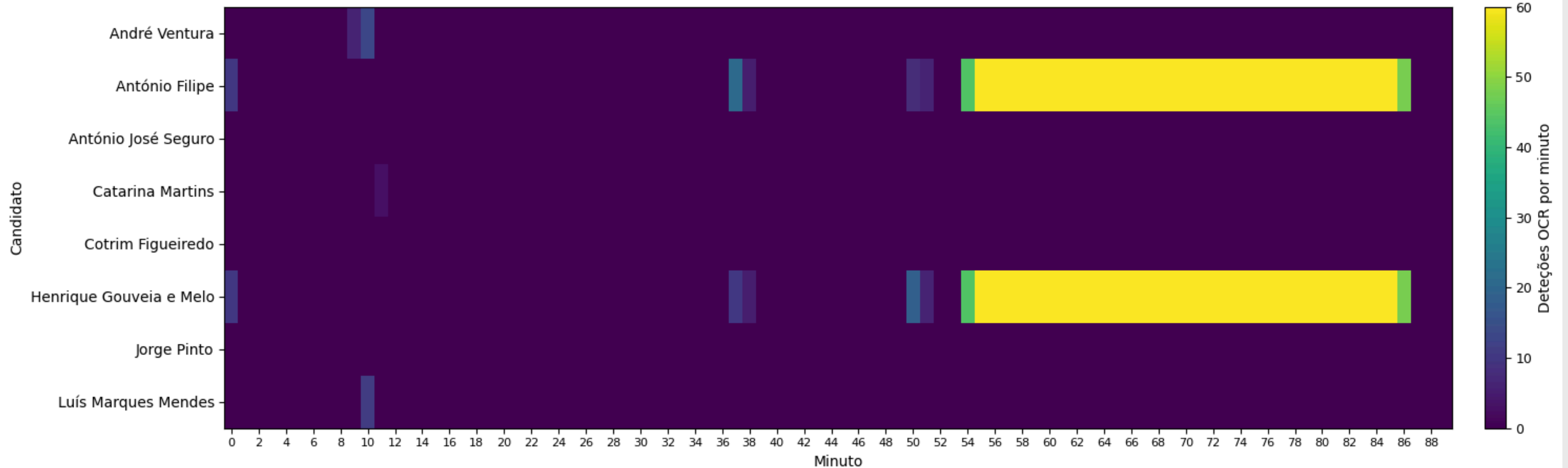




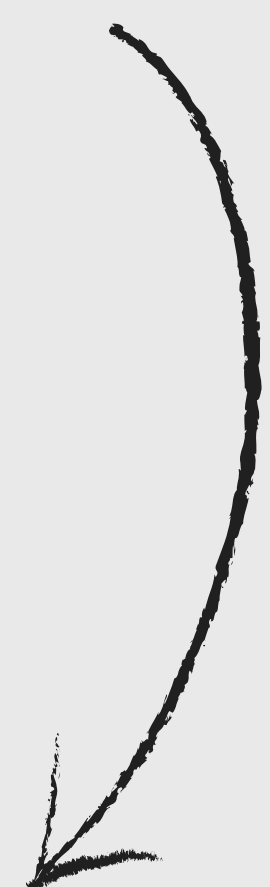
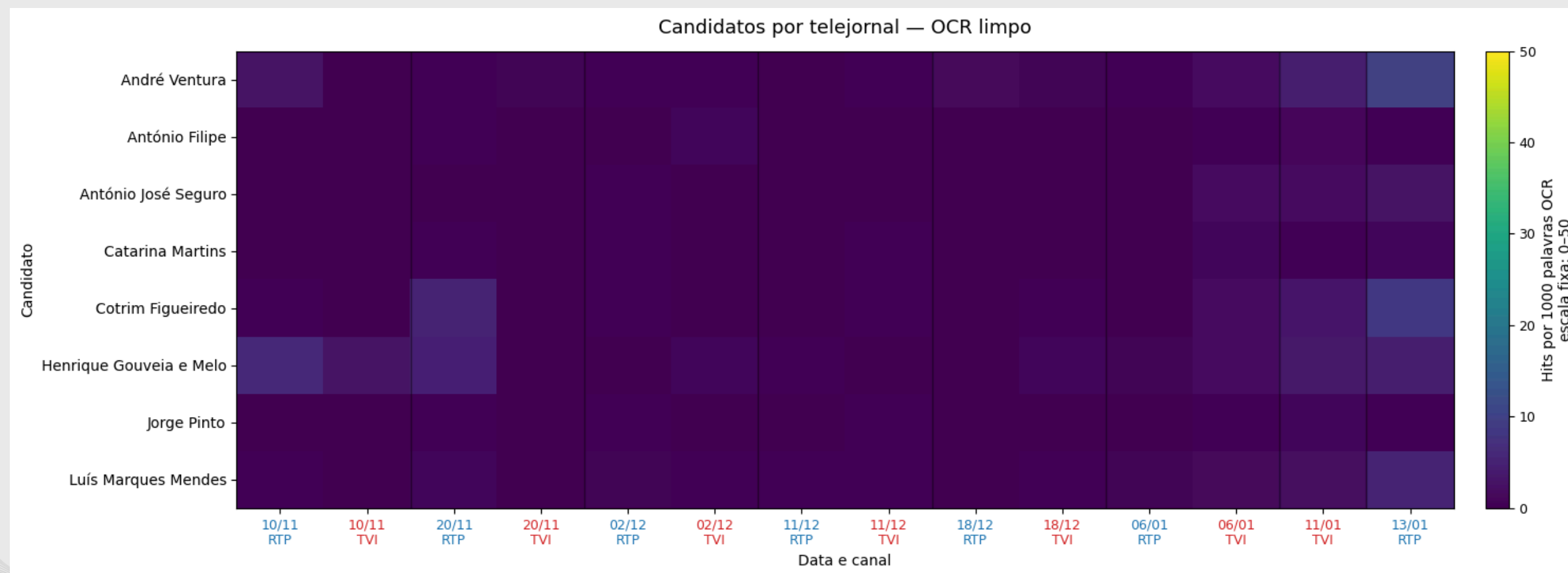
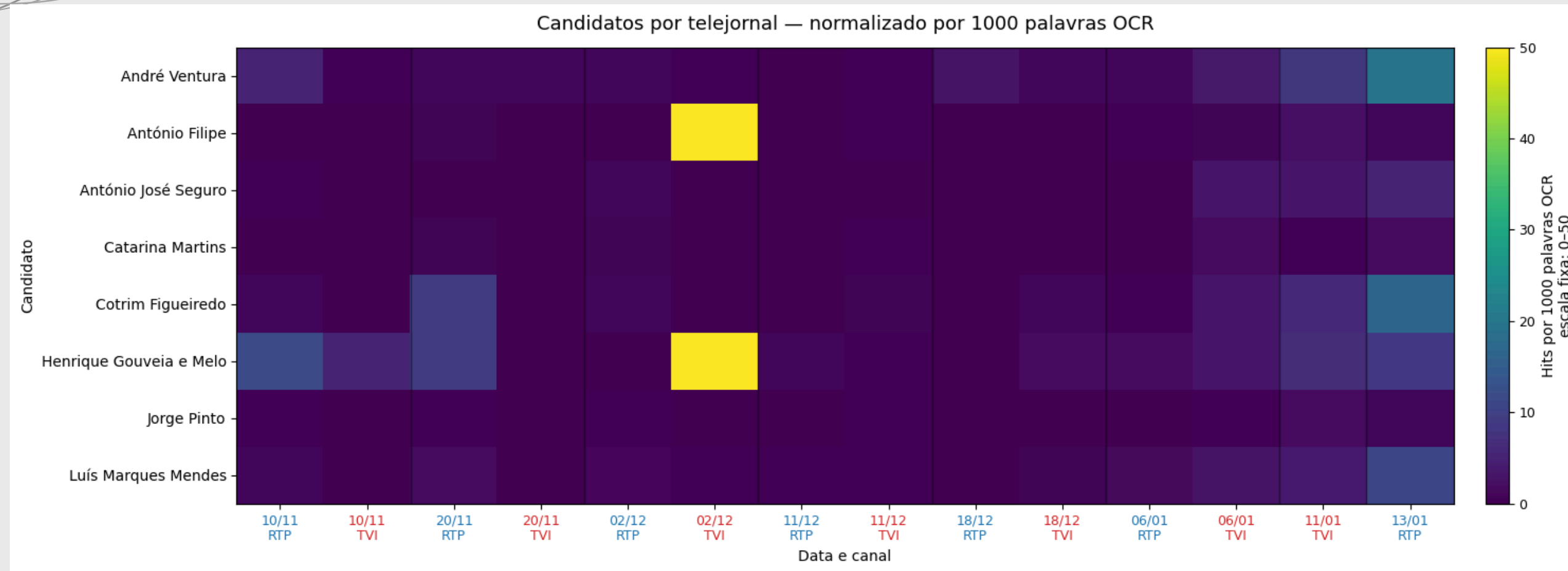
TELEJORNAL_TVI_DEC_2



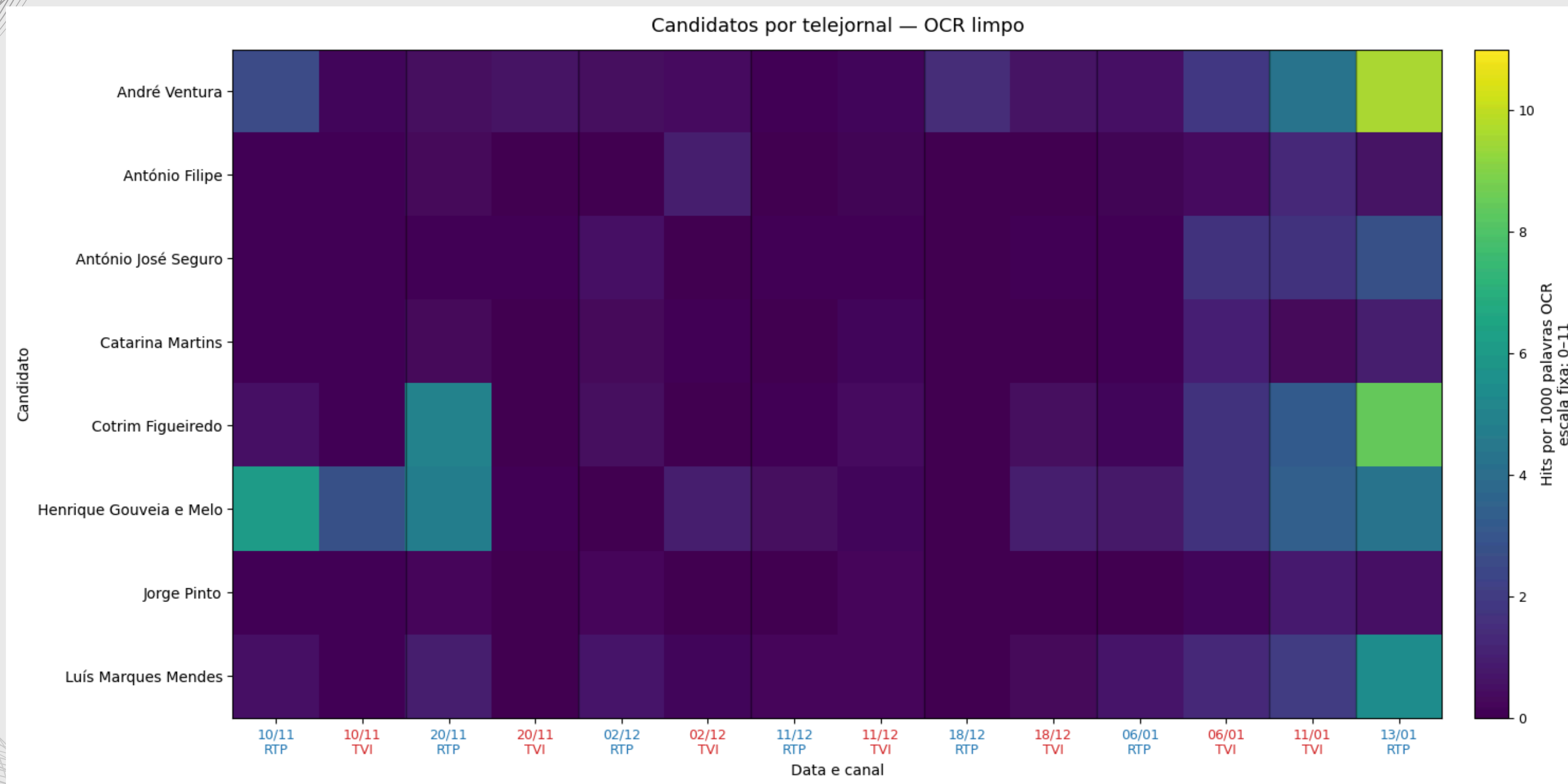
Presença de candidatos no OCR por minuto — Telejornal_TVI_Dec_2_ocr.pkl



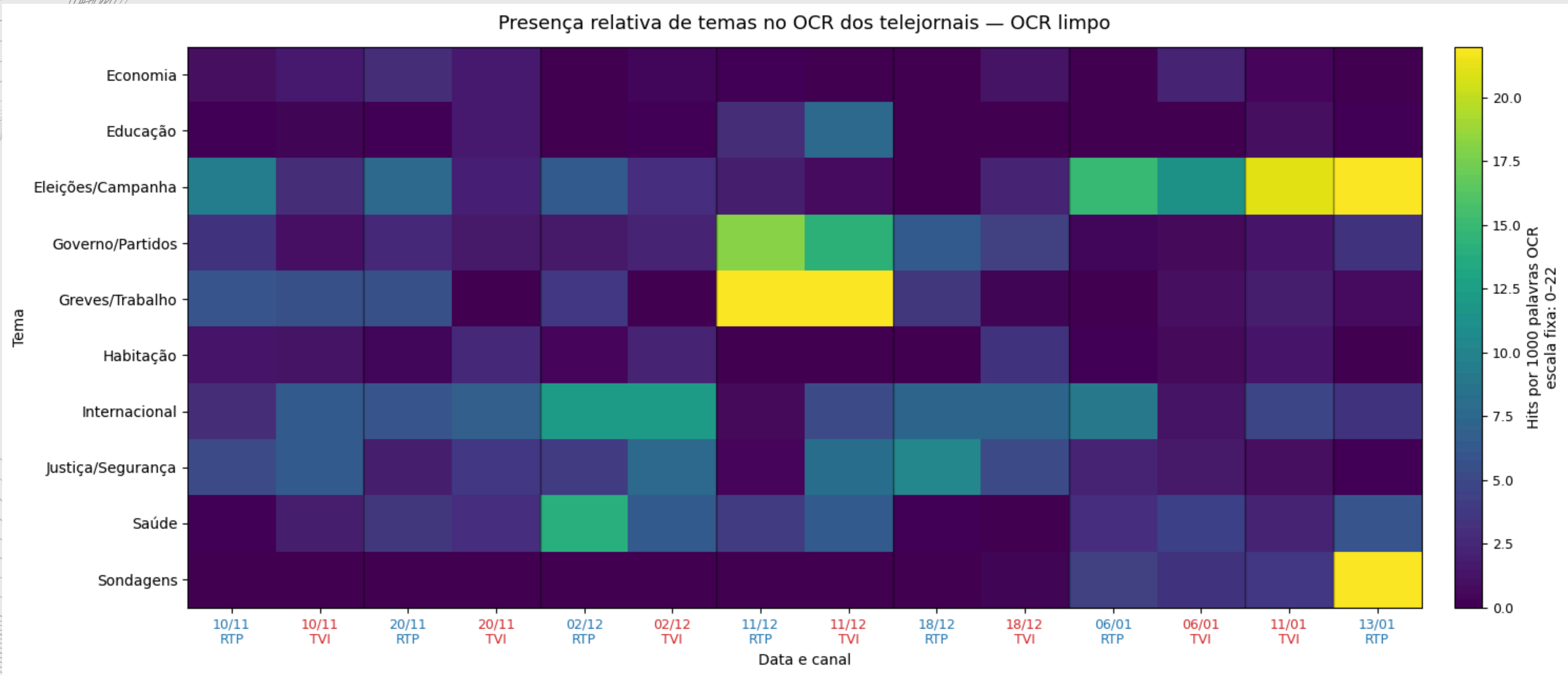
TELEJORNAL_TVI_DEC_2



TELEJORNAL_TVI_DEC_2



EVOLUÇÃO DOS TEMAS NO OCR DOS TELEJORNAIS



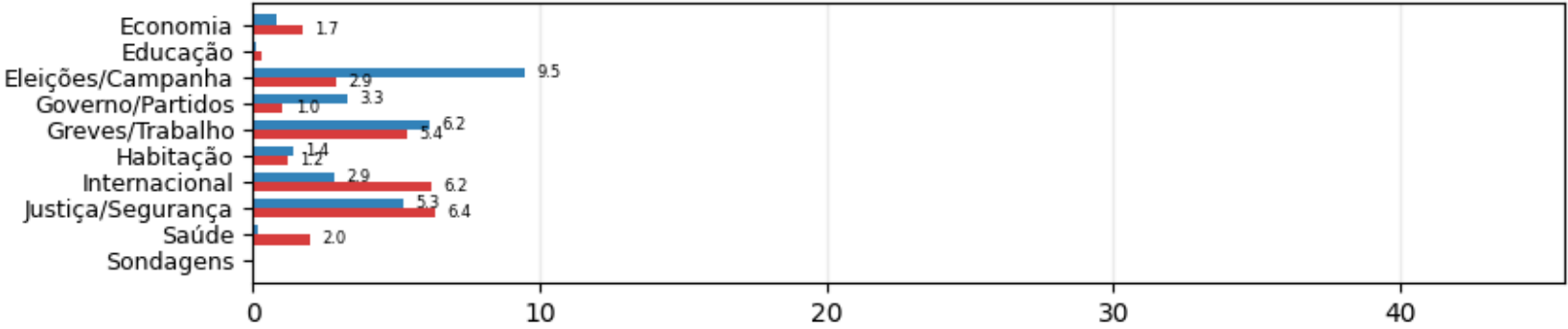
RTP VS TVI NO MESMO DIA

RTP vs TVI — comparação temática por data

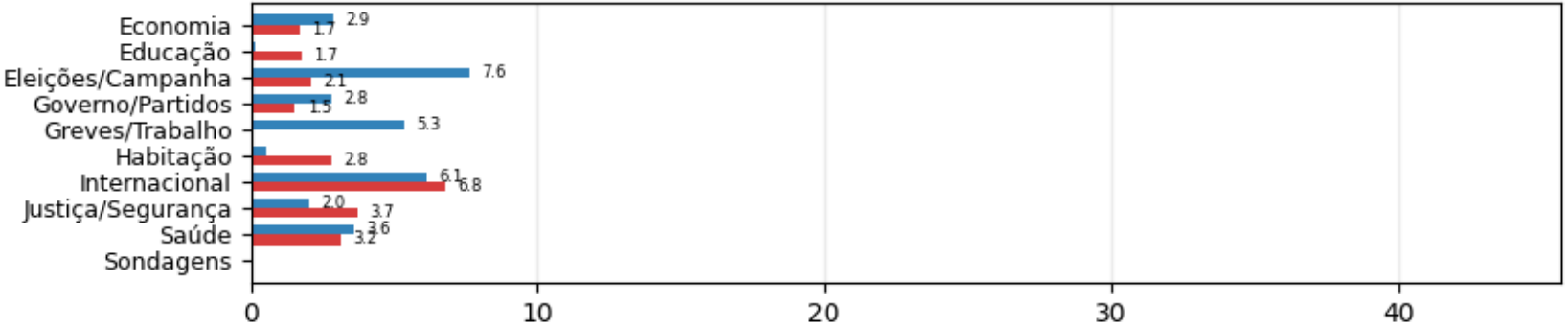
RTP TVI

Tema

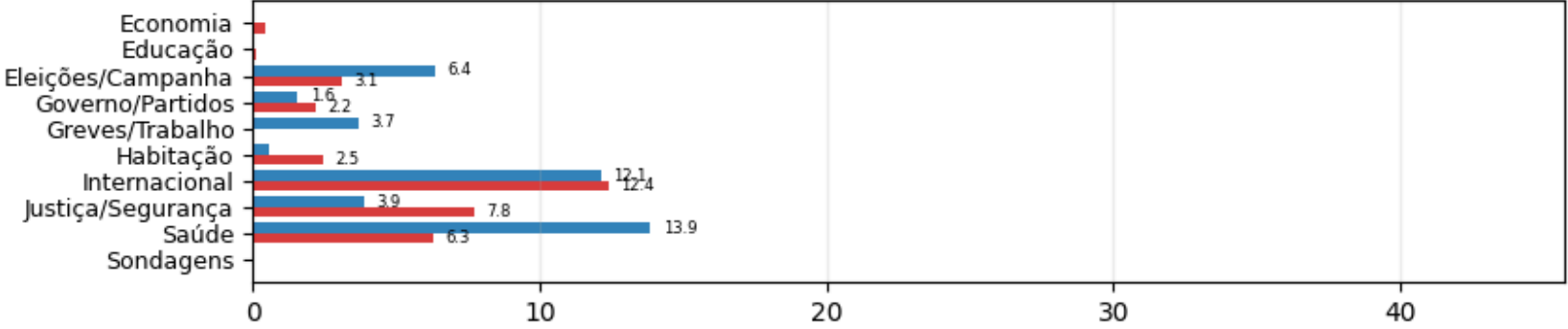
10/11



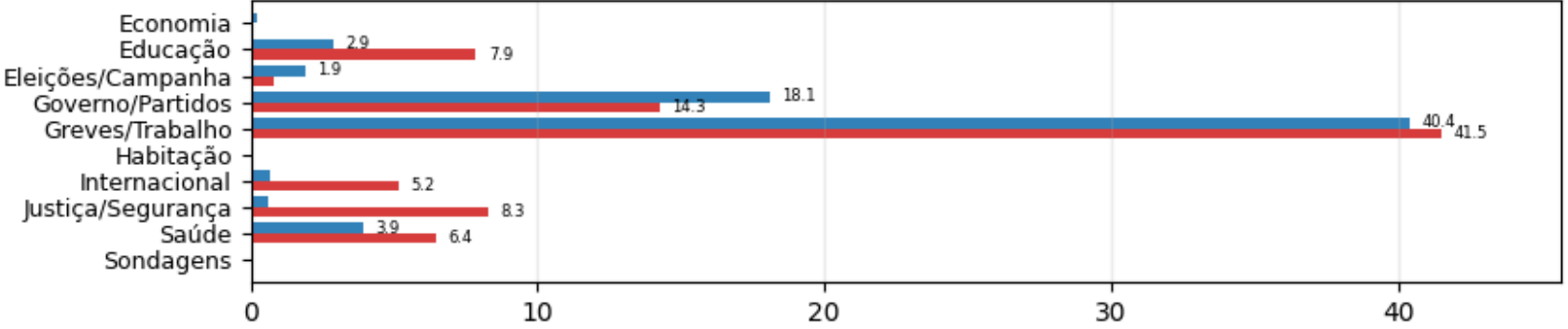
20/11



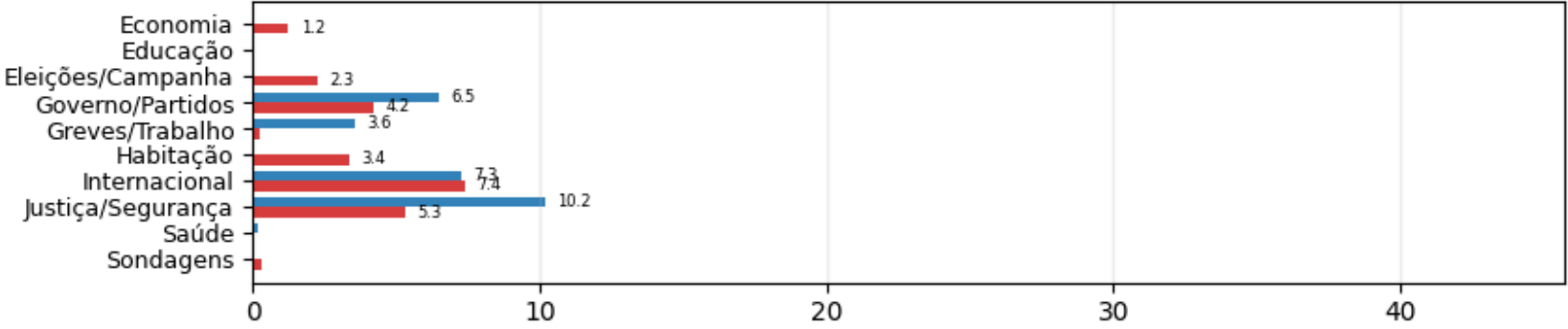
02/12



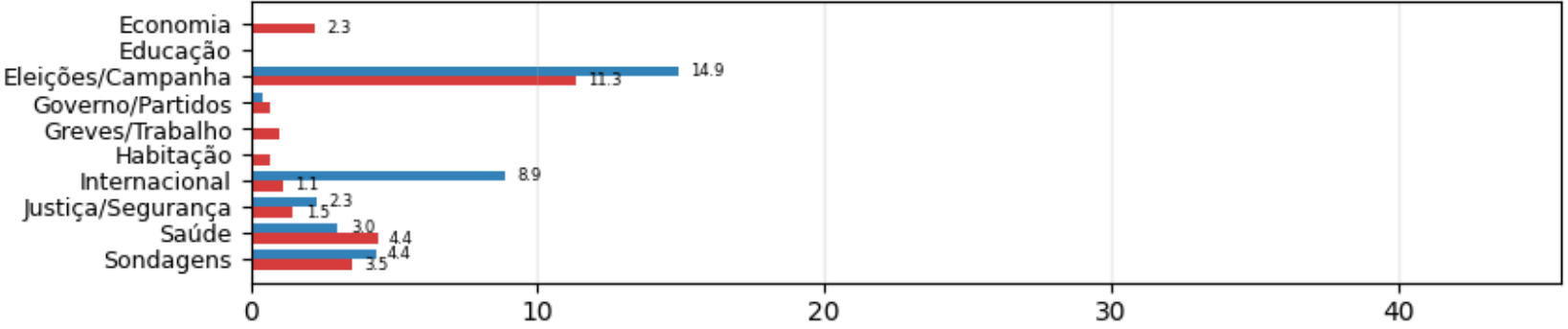
11/12



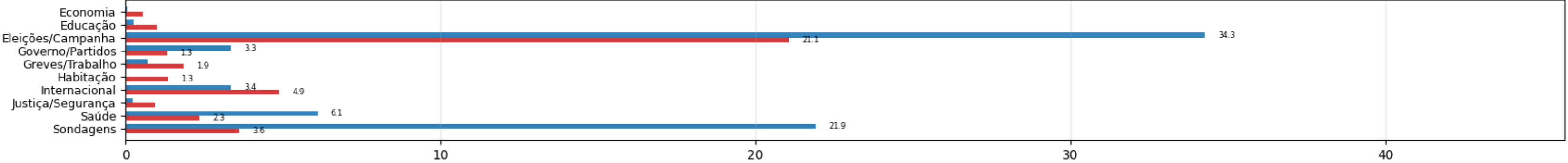
18/12



06/01



11/01 TVI vs 13/01 RTP



Hits por 1000 palavras OCR



FIM